



Infrastructure vs. Operational Intelligence

An Analytical Report Generated by the AI Research Council

Transform Airports — AI Research Council

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Abstract

Airports are over-investing in infrastructure and under-investing in operational intelligence. The next billion dollars of CapEx at major US hubs should be pressure-tested against a much smaller bet on software, data, and coordination systems that make existing assets meaningfully more productive.

How This Report Was Produced

This report is the product of the Transform Airports AI Research Council — a multi-agent analytical system running on Anthropic's Claude models. Rather than a single model writing in one pass, the Council convenes a panel of specialized research agents, subjects their work to adversarial review, and verifies every claim before publication. The process runs in four stages, with human review at two checkpoints.

Stage 1 — Independent research. Each research agent below produced an evidence brief without seeing the others' work, so the analysis draws on genuinely independent lines of evidence rather than a single line of reasoning.

Stage 2 — Synthesis and adversarial review. A Strategist synthesized the briefs into an argument; a Red Team attacked it across two rounds; the Strategist revised after each.

Stage 3 — Edit and fact-check. An Editor tightened the prose. A Fact-checker verified every number against the research briefs, removing or flagging what it could not source.

Stage 4 — Production. The verified draft was formatted into this document.

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Process agents (on every report)

- **Strategist** — synthesized the research briefs into a single argumentative draft.
- **Red Team** — attacked each draft to surface weak claims and logical gaps.
- **Editor** — tightened the prose and removed filler without adding new content.
- **Fact-checker** — verified every numerical and attributed claim against the briefs.

Each agent is governed by a versioned specification — its analytical mandate, evidentiary standards, source discipline, and argumentative posture. The complete technical definitions for every agent on the Council are published and openly available at:

<https://github.com/TransformAirports/ai-council/tree/main/.claude/agents>

Infrastructure vs. Intelligence: Why the Next Billion in Airport CapEx Deserves a Harder Test

Draft v3 — Final (Fact-Checked) Date: April 17, 2026 **Run:** infrastructure-vs-intelligence

Analyst-derived envelopes and proposed defeat-test mechanisms are tagged `[UNVERIFIED — HUMAN REVIEW]` where they sit outside Stage 1 brief evidence. Each tag marks a figure or mechanism that is analyst judgment, not brief-sourced. See `fact-check-report.md` for the full verification log.

1. Thesis

US airports are spending against a physical problem they diagnosed two decades ago, while a material share of the binding throughput constraint has migrated to coordination, staffing, and software. ACI World reports 6.3% global airport ROIC in 2024 against a weighted average cost of capital it concedes is higher. That sits alongside an 86% cross-sector megaproject overrun baseline and a PFC that has lost 40% of its construction-cost purchasing power since 2001. Part of the current capital program does not earn its cost of capital; a separate part is mandatory safety and asset-renewal work no rebalancing will touch.

The thesis is narrow: the discretionary portion of the next billion in airport CapEx — the portion not driven by pavement PCI, safety compliance, or irreducible physical overload — should have to clear a defeat test. That test is whether a much smaller operational intelligence investment, applied first, could compress apparent demand for the physical project enough to stage, defer, or resize it. Most US hubs do not run that test. Section 6 names the MWAA sub-projects that would be deferred or resized under which outcomes.

2. Executive Summary

1. **Airport CapEx, in aggregate, does not earn its cost of capital.** ACI World reports 2024 global airport ROIC at 6.3% and states directly that this remains below the sector's WACC [Economist brief, Finding 1, Quotable 1]. The precise figure comes from press releases rather than the paywalled primary report and could not be independently verified [Economist brief, Section 3, Caveats]. The direction is clear; the decimal is not. A sector that cannot clear WACC in aggregate cannot average its way out — it can only outperform with projects earning materially above the mean.
2. **Megaproject overruns are the baseline.** Flyvbjerg's 258-project transport dataset finds an 86% overrun rate at 28% average magnitude [Economist brief, Finding 2]. Airport cases run higher: Berlin Brandenburg 132% over and nine years late; LAX APM 76% over with no opening date as of March 2026; Denver Great Hall roughly three times original city exposure [Economist brief, Finding 2, Evidence]. Heathrow T5 is the counter-case; its lesson is governance, not category feasibility.

3. **The US delay picture is not primarily a runway picture.** FAA OPSNET data attribute 74% of system delays to weather and 15% to volume; BTS flight-level data attribute 39% of delayed minutes to late-arriving aircraft and 35% to air carrier causes [Operations brief, Findings 1 and 2]. Nineteen of the 30 largest ATC facilities operate below 85% of staffing targets and produce 40% of all NAS delays [Operations brief, Finding 2, citing Brookings]. These are current attributions; they do not refute the Contrarian's forecast that future demand growth (1.28B US enplanements by 2038) may be physical. Section 5.3 separates the two.
4. **Operational interventions have delivered independently validated double-digit gains.** RECAT at Memphis: 15%+ FAA-estimated capacity increase, nine added movements per hour [Operations brief, Finding 4]. TBS at Heathrow: 62% first-year reduction in headwind delays, 2.9 additional landings per hour in strong wind [Operations brief, Finding 4]. EUROCONTROL's A-CDM assessment across 17 airports: 34,000 tonnes of fuel saved annually and 250,000 minutes of flow delay avoided [Technology Scout brief, Finding 4]. These sit a category apart from vendor claims in credibility and from 40%+ marketing figures in magnitude — mid-single-digit to low-double-digit, measured, audited.
5. **Most runway expansions do not hold their value — but the 15 and 19 groups are distinct tests on the same sample.** Bilotkach/Mueller analyzed 55 runway expansions at the 150 busiest global airports, 2000–2016: 15 airports were operating peak hours below 90% of pre-expansion capacity by 2016 (expansion unnecessary on its own terms); 19 were operating at 90%+ of post-expansion capacity within a decade (construction bought only temporary relief) [Operations brief, Finding 5]. The Operations brief does not state mutual exclusivity. The honest claim is 15 of 55 unnecessary and 19 of 55 re-congested. V2's combined "34 of 55 / 62%" headline is withdrawn.
6. **MWAA's financial envelope is tightening.** The 2025–2039 capital program is approximately 5.6x the prior cycle [Economist brief, Evidence: MWAA-Specific]. Part is catch-up on deferred asset renewal; the remaining discretionary portion is this paper's subject. 2025–2028 debt issuance: approximately \$5.5 billion; adjusted debt per O&D enplanement projected to peak near \$400 (from \$223 in 2024); net revenue DSCR projected to fall from 1.63x to approximately 1.3x [Economist brief, Finding 5]. Moody's shifted the US airport sector outlook to negative in May 2025 [Economist brief, Finding 5].
7. **Operational intelligence is optionable. Concrete is not.** A bad A-CDM deployment is recoverable in tens of millions and a few years. A bad concourse is a balance-sheet obligation for thirty. That asymmetry is the strongest argument for sequencing: commit operational intelligence first and you learn what physical investment the higher operational baseline actually requires. Commit infrastructure first and you foreclose the test.
8. **What this paper asks MWAA to do, in one sentence.** Identify the discretionary sub-tranches of Project Journey and Concourse E (Sections 6.1 and 6.2), commit to an operational intelligence program sized against Technology Scout component benchmarks (Section 6.3), and pre-register that named sub-tranches are

deferred or resized if named KPIs move by named amounts by named dates (Sections 6.1, 6.2, 6.5).

3. The Argument

3.1 The economics don't work at the top of the stack

Start at the industry's own income statement. ACI World reports 2024 global airport ROIC at 6.3% and states in its own press release that this "remains below the sector's weighted average cost of capital" [Economist brief, Quotable 1]. The figure comes from a press release summarizing a paywalled report, and the 2023 ROIC and regional breakdowns could not be independently verified [Economist brief, Section 3, Caveats]. The direction of the gap is what this paper relies on; the exact percentage point is not. A trade body does not publicly concede value destruction unless the pattern is directionally robust.

Total 2024 global airport revenue of \$194.9 billion ran 2.1% below pre-pandemic levels in real terms even though passenger traffic was 4% above 2019 [Economist brief, Evidence: Global CapEx Trajectory]. Traffic recovered. Revenue did not. The explanation is structural: airports raised capital aggressively during the pre-pandemic boom and again to bridge the pandemic gap, and interest expense rose 18% year-over-year in 2023 as debt repriced against higher rates. The 2023 global airport capital cost of \$40 billion is 1% above pre-pandemic levels. ACI's long-run outlook with Oxford Economics projects \$2.4 trillion in global airport CapEx from 2021 to 2040 [Economist brief, Evidence]. That trajectory presupposes the industry can clear its cost of capital. Current evidence says it cannot.

When aggregate ROIC sits below WACC, you cannot average your way out. You outperform only with projects earning materially above the mean — and the execution margin is small.

3.2 Overruns are the baseline

Flyvbjerg's 258-project transport dataset finds 86% of projects overrun at 28% average magnitude [Economist brief, Finding 2]. Airport megaprojects run higher:

- Berlin Brandenburg: €2.8 billion budget, €6.0–6.5 billion final cost, nine years late, total value destruction near €11.9 billion including interest and unrealized revenue [Economist brief, Evidence: Cost Overrun Case Studies].
- LAX APM: \$1.9 billion contract in 2018, \$3.34 billion as of July 2024, no opening date as of March 2026 [Economist brief, Evidence].
- Denver Great Hall: approximately \$770 million original city exposure, now \$2.1–2.35 billion across phases. A January 2025 city audit found DIA "still is not following procurement procedures or requiring contractors to submit documentation of actual costs" [Economist brief, Evidence].

- Denver original automated baggage system (1995): \$560 million in overruns, deployed to one concourse for one airline's outbound flights [Economist brief, Evidence].

GAO-20-298 makes the procurement problem concrete: airport planned infrastructure costs for FY2019–2023 averaged \$22 billion annually, a 19% increase over prior estimates, with terminal projects up 51% [Economist brief, Evidence: US Federal Framework]. Not inflation. Optimism bias embedded in planning.

Heathrow T5 is the counter-case. The £4.3 billion terminal opened on schedule in March 2008, within budget, after BAA ran a two-year pre-construction study designed to avoid megaproject failure modes [Economist brief, Evidence: Cost Overrun Case Studies]. The lesson is not that the sector hits targets as a matter of course; it is that hitting targets required pre-commitment to governance most authorities do not make. Section 6.4 specifies what "T5-class governance" means operationally.

3.3 The delay data does not say what the planning data assumes

Walk into a US airport planning meeting and you hear about runway throughput, gate counts, and peak-hour movements. Walk over to the FAA's delay data and the picture shifts.

FAA OPSNET data over the six years ending May 2023 attribute 74.26% of system-impacting delays (>15 min) to weather, 14.92% to volume, 5.96% to runway unavailability, 0.61% to equipment, and 4.25% to other factors [Operations brief, Finding 1]. Read quickly, nothing looks tractable: weather is weather. Read carefully, "volume" delays are not physical runway limits — they are the gap between instantaneous demand and the *operationally realized* capacity of an airport at that moment. That gap is where coordination lives.

BTS flight-level data records what airlines experience rather than what the NAS produces. From January–April 2023, late-arriving aircraft accounted for 38.99% of total delayed minutes, air carrier delays 35.43%, and NAS delays a smaller share [Operations brief, Finding 2]. Late-arriving-aircraft delay is a scheduling and turnaround phenomenon. It has no physical solution. You cannot concrete your way out of an aircraft that landed at LGA 20 minutes behind and now has to push back late at EWR.

ATC data completes the picture. The FAA has approximately 11,000 certified professional controllers against a model requirement near 14,000 [Operations brief, Finding 6]. Nineteen of the 30 largest facilities operate below 85% of staffing targets and account for 40% of all NAS delays [Operations brief, Finding 2, citing Brookings]. At peak dysfunction, DOT Secretary Duffy reported controller staffing accounted for half of all flight delays versus a normal 5% baseline [Operations brief, Finding 6].

None of this changes with a new concourse. The Contrarian's own Scenario 2 concedes: "At some airports, delay is genuinely a coordination problem rather than a physical capacity problem... The argument for operational intelligence is strongest where this

diagnosis is correct and infrastructure investment is therefore treating a symptom rather than a cause" [Contrarian brief, Part III, Scenario 2].

This speaks only to current delay attribution. Whether *future* demand growth is physical is addressed in 3.6 and 5.3.

3.4 The operational-intelligence evidence base is real, not theoretical

The honest version cites only data points that survive scrutiny.

RECAT at Memphis (2012). The FAA reclassified aircraft wake-turbulence categories. No construction. Result: 15%+ FAA-estimated capacity increase, nine additional movements per hour, FedEx reporting 4.18 million gallons of fuel saved annually, monthly fuel savings of \$1.8 million, and a three-minute cut in average taxi time [Operations brief, Finding 4]. Months, not a decade of review.

Time-Based Separation at Heathrow (2015). NATS replaced distance-based arrival separation with time-based, neutralizing the headwind penalty that collapses runway throughput in strong winds. First year: 62% reduction in headwind delays, 2.9 additional landings per hour in strong wind, more than 80% of landing aircraft separated more closely than under the old system [Operations brief, Finding 4]. Over ten years: 1.5 million minutes of airborne holding removed, 100,000+ tonnes of fuel saved, approximately USD 70 million in fuel cost savings [Operations brief, Evidence].

Honest framing: Heathrow operates at 98% of its 480,000 annual movement cap [Operations brief, Counterexamples]. TBS delivered large throughput gains *while the third-runway debate proceeded in parallel* — it did not substitute for the runway. Heathrow is still pursuing the third runway because TBS's gains were insufficient to close the demand gap at a two-runway airport serving 1,300+ daily movements. For airports not at a 98%-of-cap ceiling, TBS-class interventions can plausibly defer or resize the infrastructure question. For Heathrow itself, they did not.

A-CDM across 17 European airports — and the network-effect caveat.

EUROCONTROL's independent 2016 impact assessment — from network operations data, not vendor-reported — documented 34,000 tonnes of annual fuel savings (~€26.7M at then-current prices) and approximately 250,000 minutes of flow delay avoided (~€15.5M) [Technology Scout brief, Finding 4]. Two caveats: (1) "the marginal benefit of adding one more airport decreases as the network penetration increases" — European A-CDM gains are partly a network effect a US deployment will not fully capture [Technology Scout brief, Finding 4]; (2) the FAA's TFMS is not a CDM equivalent, so a US A-CDM deployment at a single airport is more isolated than the European benchmarks imply [Technology Scout brief, Section 4].

The caveat has to flow through to the MWAA recommendation, not just be transmitted. Section 6.2 resolves this by recommending A-CDM at *both* DCA and IAD as a paired regional deployment; sizing the envelope against Munich's single-airport benchmark rather than the 17-airport network benchmark; and setting the defeat-test threshold

below Munich's figures. Munich sat inside the European network, so it is itself an inexact proxy — but it is the closest single-airport anchor the brief evidence provides.

Munich's SESAR DMAN trials produced granular airport-level numbers: 9% taxi time reduction, 14.6 kg fuel savings per flight, ATFM slot adherence improved from 76% to 81%, off-block time predictability up 7.8% [Technology Scout brief, Finding 4].

FAA TFDM at Charlotte. Four years of testing saved more than 275,000 gallons of fuel per year, cut GHG emissions by 8 tonnes CO₂ daily, and eliminated 916 hours of delay per year at a single airport [Operations brief, Evidence: FAA TFDM Surface Metering]. At national rollout, the FAA projects 7 million gallons saved annually and 75,000 tonnes of CO₂ eliminated.

Single-digit to mid-double-digit percentage improvements — smaller than vendor marketing, larger than zero, and large enough to change a capital plan if taken seriously.

3.5 The price-tag ratio, honestly framed

Global airport IT spending reached \$8.9 billion in 2024; global airport capital costs in 2023 were approximately \$40 billion [Economist brief, Finding 6]. Gross ratio: roughly 1:4.5. But SITA's 2024 Air Transport IT Insights survey identifies cybersecurity as the #1 IT priority at 100% of respondents, with passenger processing #2 at 95% [Technology Scout brief, Evidence: Industry IT Spending Context]. The overwhelming share of that \$8.9 billion is not buying A-CDM or DMAN or surface metering — it buys cybersecurity, passenger processing, payroll, networking, and back-office systems.

Operational intelligence in the narrow Technology Scout taxonomy — A-CDM/DMAN, predictive turnaround/ramp coordination, operational-simulation digital twins, AI queue prediction, integrated operations centers — is a narrow slice. Deployment counts give a floor: only 3 of 31 US large hubs have operational digital twins [Technology Scout brief, Finding 2]; A-CDM is live at 34 *European* airports with no US equivalent at comparable scale [Technology Scout brief, Finding 4]; SFO's AIOC opened in January 2026 as one of the first serious US APOC deployments [Technology Scout brief, Section 4]. Narrow-sense operational intelligence is at most a single-digit-percentage-of-IT-spend category.

That sharpens the allocation argument. The ratio is not \$8.9B of operational intelligence against \$40B of concrete. It is much less operational intelligence against \$40B of concrete. V1's "closer to 1:10 or worse at many large US hubs" is not in any Stage 1 brief and remains removed.

- Gross: airport IT of \$8.9B vs. airport CapEx of \$40B = 1:4.5 globally.
- Net: the narrow "operational intelligence" subset is a small fraction of \$8.9B; against \$40B of CapEx, a much larger asymmetry.
- Per project: the LAX APM alone at \$3.34 billion consumes more than a third of annual global airport IT spend [Economist brief, Evidence]. The upper-range estimate for a large-hub digital twin (\$15–40M over 24–36 months plus 10–20%

annual recurring) [Technology Scout brief, Evidence: Digital Twins] is roughly 1.2% of the APM's current cost overrun alone.

The argument does not turn on decimal precision. It turns on the order of magnitude between physical CapEx and the operational intelligence subset. That order of magnitude holds.

3.6 Induced demand, welfare, and capacity that evaporates

The Bilotkach/Mueller study of 55 runway expansions at the world's 150 busiest airports, 2000–2016, reports two separate findings on the same population. Fifteen were operating peak hours below 90% of pre-expansion capacity — expansion unnecessary on their own terms. Nineteen were at 90%+ of post-expansion capacity within a decade [Operations brief, Finding 5]. The brief does not state mutual exclusivity; v2's combined "34 of 55 (62%)" headline assumed non-overlap without sourcing and is withdrawn. The two findings stand on their own.

The induced-demand challenge is fair: re-congestion after expansion is not automatically a failure. If new capacity absorbed latent demand previously rationed, that is welfare-positive — the airport now serves passengers it previously could not. Denver at 82.4 million through a 50-million-design airport is the canonical example [Contrarian brief, Part IV-3]. The narrow claim: expansions whose economic case depended on sustained post-expansion headroom — the justification most planning documents use to underwrite bonds — are defeated when that headroom disappears within a decade. That is the 19-of-55 pattern. Expansions in markets where the pre-expansion level itself went unreplicated — 15 of 55 — are defeated on their own planning terms regardless of welfare counterfactuals.

In the remaining cases, the welfare case may be strong but the financial case depends on demand projections that, at airport timescales, cannot be underwritten with confidence. Infrastructure pays where the binding constraint is physical and demand is inelastic over the asset's 30–50 year life. Frankfurt Runway 18 West raised movements per hour from 83 to 126 — a 52% increase — because the prior geometry did not permit simultaneous independent operations and no procedural change could replicate that [Operations brief, Counterexamples]. Legitimate case. Not the modal US hub.

The Contrarian brief is correct that FAA and IATA forecast sustained enplanement growth (1.28 billion US by 2038; 7.8 billion global by 2036) [Contrarian brief, Part I-4]. Those forecasts are not refuted by the *current* delay data in 3.3, and the thesis does not require them to be refuted. Where the infrastructure case turns on future demand that is genuinely physical — DEN at 82.4M through a 50M-design airport is the clearest case — the thesis does not apply. Sections 3.7 and 5.1 specify this scope.

3.7 MWAA's balance sheet, and naming the discretionary tranche

MWAA's 2025–2039 Capital Construction Program is valued at approximately \$9 billion, replacing a 2015–2024 program of roughly \$1.6 billion — a 5.6x step-up [Economist brief, Evidence: MWAA-Specific]. The bare 5.6x assumes the prior cycle was

correctly sized. It was not, at least in part: US airport infrastructure has been running \$142B behind projected 10-year need [Economist brief, Finding 3], and the Contrarian's asset-depreciation and FAA PCI-driven pavement argument [Contrarian brief, Part II] applies to any multi-decade plan. Part of the 5.6x is genuine catch-up on deferred safety, security, accessibility, and asset-renewal work. Not what this paper is about.

Sizing the discretionary portion — order of magnitude, not decimals. Neither Stage 1 brief computes the figure. The Economist brief identifies the \$9B envelope, names the two largest projects (Project Journey at \$2.4B; Concourse E within the \$9B without a sub-project figure), and does not decompose either into mandatory vs. discretionary. The Contrarian brief states the \$151B ACI-NA need is "primarily mandatory safety, security, and accessibility compliance projects" and that "more than 50% of Airport Improvement Program funding goes toward constructing or rehabilitating runways, taxiways, and aprons" [Contrarian brief, Part II].

With those two anchors and no authority-level disclosure, the honest construction is an order-of-magnitude band:

- If the AIP 50%+ pavement/safety/compliance ratio applies to MWAA, roughly \$4.5B of the \$9B is mandatory asset renewal and compliance. A further \$1–2B is plausibly safety, security, and accessibility work regulatorily mandated and unavoidable. That leaves \$2.5–3.5B as the upper bound of the discretionary tranche.
- Under a narrower definition — demand-contingent terminal square footage, gate count beyond slot-cap requirements, passenger-experience investment not tied to safety/compliance — Project Journey (\$2.4B) and Concourse E bracket the candidates, and the discretionary portion is more likely \$1–2B.
- A conservative analyst range is therefore **\$1–3B of the \$9B** [UNVERIFIED — HUMAN REVIEW] — roughly 10–35% of the program. That is the range against which the defeat-test argument bites. Not 5%. Not 50%.

Analyst-derived from the two brief anchors; not disclosed at that granularity in any Stage 1 source. Human reviewer should verify against MWAA's ACFR, bond filings, and published capital plan before external citation. Until then, the allocation test applies to something on the order of \$1–3B of MWAA's \$9B.

The financial environment frames everything. MWAA plans to issue approximately \$5.5 billion in new debt between 2025 and 2028. Adjusted debt per O&D enplanement is projected to nearly double from \$223 in 2024 to approximately \$400 in 2028. Net revenue DSCR is projected to fall from 1.63x in 2024 to approximately 1.3x over five years — still above bond covenant minimums (typically 1.10x–1.50x), but the cushion is shrinking. At Dulles, DSCR is projected to deteriorate from 2.79x in FY2024 to 1.70x by FY2030 even under favorable traffic assumptions [Economist brief, Finding 5]. Moody's shifted its US airport sector outlook from stable to negative in May 2025 [Economist brief, Finding 5].

The PFC backdrop compounds it. The statutory cap has been \$4.50 since 2001. A RAND study commissioned by Congress concluded that \$4.50 had fallen to the equivalent of \$2.72 in year-2000 dollars indexed to construction prices — a 40% real loss — and the 2024 FAA Reauthorization left the cap unchanged [Economist brief, Finding 3]. Airports finance an expanding capital envelope from a passenger-level revenue stream whose real value is shrinking.

3.8 The opportunity cost is not recoverable on airport timescales

The deepest argument against the marginal *discretionary* dollar of airport CapEx is not that the project will fail. It is that the capital is committed for 30–50 years whether the project succeeds or not. Bond covenants require debt service regardless of utilization. A terminal built in 2026 for a demand projection that proves wrong in 2035 cannot be un-built.

The structural warning case is CVG. After Delta's dehubbing, Cincinnati's CPE rose from under \$5 to over \$10 on collapsing enplanements [Economist brief, Finding 7]. The infrastructure was fine. The airlines left. The capital stack had no flexibility.

Operational intelligence has different optionality. A-CDM, an APOC, surface metering, computer-vision turnaround monitoring run single-digit-million to low-tens-of-million, deploy in months to a few years, and reconfigure as the operating environment changes. Assaia's 2025 report on AI turnaround monitoring across 15 airports and 450,000+ turns found median departure delay cut from 4 to 3 minutes (a 25% reduction), with North American airports saving approximately \$100 per turnaround per minute of delay reduction [Technology Scout brief, Finding 7]. Caveat: vendor-published research from vendor-customer airports; not peer-reviewed, not independently audited, sampling methodology undisclosed [Technology Scout brief, Predictive Turnaround / Ramp Coordination]. Treat as mechanistically plausible, subject to aggressive discounting. The higher-confidence anchor remains Munich's SESAR DMAN: 9% taxi time reduction, 14.6 kg fuel savings per flight, disclosed methodology [Technology Scout brief, Finding 4].

A bad airport-level operational intelligence investment is recoverable in tens of millions and months; a bad concourse is a balance-sheet problem for 30 years. Software failure is not costless — NextGen proves it can cost tens of billions at federal scale, and 5.2.1 addresses that directly. At the *airport* level, under the *commercial* deployment model this paper recommends, the optionality asymmetry holds. Scoped, not universal.

3.9 The false-precision trap in on-time performance data

Over 2012–2023, average US airline schedule padding rose 27%, adding roughly 3 minutes per flight. Research cited by FinanceBuzz finds a 10% increase in scheduled block times produced a 20-percentage-point artificial improvement in measured on-time performance over five years [Operations brief, Finding 7]. Zhang, Salant, and Van Mieghem (2018) document this as a systematic competitive response to chronic operational variance.

Airport capital programs are often justified, in part, by recent OTP and passenger-experience gains. Those metrics are partially accounting. True operational variance — the gap between realized and potential airport capacity — is larger than the reported numbers suggest. Strip the padding out and the case for operational intelligence gets stronger: the hidden variance is what A-CDM, TBS, and surface metering compress.

3.10 Summary of the case

- Industry aggregate ROIC is reported below WACC by the industry's own trade body, with a verification caveat on the precise figure.
- Megaproject overruns are the baseline; terminal projects are the worst category.
- The dominant *current* delay drivers are weather resilience, ATC staffing, and carrier-side coordination — not runway or terminal constraints at the modal US hub.
- Operational interventions have delivered independently validated single-digit-to-low-double-digit capacity gains in months-to-years, at percentages of megaproject cost.
- Runway expansion fails to hold its value in 19 of 55 documented cases and was unnecessary on its own terms in 15 of 55; overlap undetermined in the briefs.
- MWAA's discretionary capital tranche is on the order of \$1–3B of the \$9B.
- MWAA's debt trajectory is moving toward a DSCR that leaves little room for a demand or rate shock.
- Misallocated discretionary airport capital locks in for decades; a misallocated airport-level operational intelligence dollar is recoverable in years.

The argument is not that airports should stop building. It is that the discretionary portion of the next billion dollars should have to prove it belongs on the physical side of the ledger, under a defeat test almost no planning framework applies.

4. The Counter-Case, Honestly Presented

The strongest version of the pro-infrastructure argument is not a defense of current practice. It is a set of claims that together describe conditions under which operational intelligence cannot substitute for physical investment. This section presents them using the Contrarian's own framings where load-bearing, with attribution.

4.1 Physical ceilings are real, and software is not geography

Runways, taxiways, gate frontage, and airspace sectors have mathematically fixed throughput maxima. LaGuardia is capped by the FAA at 71 scheduled operations per hour because its two-runway geometry produces effective single-runway operation during peak hours in a slot-saturated New York corridor [Contrarian brief, Part I-1]. DCA handles 25 million annual passengers on infrastructure designed for 15 million; the Potomac, federal facilities, and residential neighborhoods form a hard perimeter; the FAA has imposed a hard cap of 62 takeoffs or landings per hour and stated no additional flights can be accommodated [Contrarian brief, Part II, DCA section]. Denver processed 82.4 million passengers in 2024 through infrastructure built for 50 million — 65% over

design capacity — and is planning four new concourses to reach 125 million [Contrarian brief, Part I-4; Part II]. As the Contrarian puts it: "The physics do not negotiate" [Contrarian brief, Part I-4].

The software case concedes physical ceilings in principle and can quietly ignore them when costing alternatives. Not in this paper. Sections 5 and 6 segregate ceiling cases from discretionary cases explicitly.

4.2 O'Hare is the cleanest infrastructure ROI case in modern commercial aviation

The O'Hare Modernization Program — converting an intersecting-runway airfield to six parallel east-west runways — produced a documented 64% reduction in system-impact delays comparing 2003–2008 to 2009–2020, a 79% reduction in flight delay rates, and a 50% increase in peak operational capacity [Contrarian brief, Part I-2; Operations brief, Counterexamples]. These are measured FAA outcomes over a decade, not modeled projections. The mechanism cannot be replicated in software: intersecting runways force mandatory sequencing; parallel runways dissolve that dependency.

One caveat: O'Hare's expansion also included operational and technology upgrades (Terminal Sequencing & Spacing, Time-Based Flow Management), making it difficult to isolate the pure infrastructure contribution [Operations brief, Counterexamples]. That does not overturn the core case. It does suggest that even the cleanest infrastructure ROI story in modern aviation was executed as a coupled infrastructure-plus-operations program — which is how 6.1 and 6.2 recommend MWAA treat its own major projects.

4.3 NextGen is the empirical indictment

This is the single most damaging data point in the debate. The FAA's NextGen program — the largest operational intelligence initiative in the history of commercial aviation — has consumed over \$15 billion in federal investment (combined public and private through 2030 exceeding \$35 billion) and, per the DOT OIG, delivered approximately 16% of projected benefits [Contrarian brief, Part I-5]. The original \$199 billion 2013 projection was revised to \$63 billion by 2024. The NAS Voice System was terminated after \$160 million with minimal benefits, then required an additional \$274 million to sustain legacy systems through 2030. The Terminal Flight Data Manager program was delayed three years, cost 20% more than budgeted, and had its scope cut from 89 to 49 sites [Contrarian brief, Part I-5; Part II].

The DOT OIG is direct: the program is "narrower in scope, more expensive, and later than the version originally promised" [Contrarian brief, Part I-5]. Benefit-to-cost ratio for 2010–2018 was below 1.0 [Contrarian brief, Part I-5].

Any proposal to reallocate discretionary airport capital to operational intelligence has to explain why the largest operational intelligence program in aviation history delivered a BCR below one, and what is structurally different about the investments this paper proposes. Section 5.2.1 answers directly.

4.4 The base rate for AI project success in infrastructure and operations is low

A Gartner analysis published in April 2026 found that only 28% of AI projects in infrastructure and operations fully succeed and meet expectations; 20% fail outright [Contrarian brief, Part I-7]. Among leaders reporting failure, 57% cited "expecting too much, too fast." Data quality issues and skill gaps are the other leading causes — and aviation data is particularly hostile on both: inconsistent formats across carriers and ground handlers, legacy system encoding, real-time latency, and no common data standard across stakeholders [Contrarian brief, Part I-7].

Between 60–85% of enterprise AI projects fail overall, with 68% of failures attributable to data quality issues [Technology Scout brief, Finding 5]. SITA's 2024 survey identifies fragmented data as the primary barrier to realizing value from \$8.9 billion in annual airport IT spending [Technology Scout brief, Finding 5]. SEA spent \$13.4 million integrating seven vendors (Saab Aerobahn, Assaia, Inform GroundStar, Safedock, Genetec, Accipiter, FlightAware) for basic ramp situational awareness, and the case study reports no quantified outcome — secondary-source figures of "17% on-time performance improvement and 44% taxi-in reduction" could not be confirmed to a primary source [Technology Scout brief, Quotable 3, Finding 1]. Section 5.2.2 addresses SEA directly.

4.5 Infrastructure is accountable; operational intelligence, characteristically, is not

A runway, a concourse, a gate — inspectable, auditable, observable. When they fail, everyone sees it. When a \$15 billion operational intelligence program delivers 16% of promised benefits, the vendor community's response is not to refund but to argue for more funding and longer timelines [Contrarian brief, Part I-8].

The counter-response in Section 5.2.4: Denver Great Hall, LAX APM, and Berlin Brandenburg were all inspectable, auditable, and still catastrophic. The accountability argument is an argument for governance structure, not for concrete over code.

4.6 Software creates new fragility; infrastructure creates durable redundancy

The July 2024 CrowdStrike incident grounded Delta for multiple days. A single failed hardware component at a data center grounded Alaska Airlines for hours in 2025, producing over 150 cancellations [Contrarian brief, Part I-6]. The GAO found that 40 of 41 airports it surveyed are actively investing in physical electrical resilience — backup power, microgrids, redundant distribution — because operational intelligence requires physical substrate to function [Contrarian brief, Part I-6]. A runway does not crash. Concrete does not experience an unplanned outage.

4.7 Demand is real and growing faster than optimization can absorb

The FAA's 2025–2045 Aerospace Forecast projects US enplanements surpassing 1.28 billion annually by 2038. IATA projects global air travel reaching 7.8 billion passengers by 2036 [Contrarian brief, Part I-4]. Operational intelligence can improve throughput at

the margin. It cannot serve 82 million passengers through a terminal designed for 50 million. This paper concedes the forecast. Section 5.3 separates the forecast-is-physical question from the current-delay-attribution question rather than conflating them.

4.8 The \$151 billion backlog is largely non-discretionary

ACI-NA's 2023 Infrastructure Needs Report puts US airport infrastructure need at \$151 billion over five years, up more than 30% from \$115 billion two years prior [Contrarian brief, Part I-3; Part II]. Contrarian Part II: "the \$151 billion ACI-NA infrastructure needs figure... reflects primarily mandatory safety, security, and accessibility compliance projects, not discretionary amenity investment" [Contrarian brief, Part II, Asset Depreciation]. More than 50% of AIP funding goes to constructing or rehabilitating runways, taxiways, and aprons because FAA safety standards require it. A runway with a deteriorated PCI score must be rehabilitated before other capital projects get AIP approval [Contrarian brief, Part II].

This is the single most important concession the thesis makes, and Section 5.1 concedes it directly. The allocation argument applies to discretionary capital, not mandatory compliance and asset-renewal. Section 3.7 quantifies what fraction of MWAA's \$9B that leaves in scope.

4.9 Synthesizing the counter-case

Put together, the pro-infrastructure argument is:

9. Physical ceilings exist and are absolute where they bind.
10. Where the ceiling is genuinely physical, infrastructure has a measured track record of large and durable gains (O'Hare, Frankfurt).
11. The largest operational intelligence experiment in aviation history delivered 16% of its promises.
12. Base rates for AI success in infrastructure and operations are low; the aviation data environment is harder than average.
13. A majority of the \$151B need is mandatory safety and asset-depreciation investment.
14. Demand growth is forecast to be sustained.

That is a serious argument. It deserves a serious answer.

5. Why the Counter-Case Is Insufficient

The counter-case has to be conceded where right. The thesis still holds on a narrower footprint. This section leads with the strongest rebuttal (NextGen), then SEA (the strongest US airport-level counterexample), then the Gartner base rate (engaged rather than defined away), then softer concessions.

5.1 What must be conceded

Physical ceilings are real. The thesis does not survive an airport operating at 98% of a genuine physical cap. Heathrow qualifies [Operations brief, Counterexamples]. DCA is at a federal slot cap plus physical perimeter [Contrarian brief, Part II, DCA section]. LGA

is at a hard FAA cap with geometry that enforces effective single-runway operation at peak [Contrarian brief, Part I-1; Part II]. Where the binding constraint is physical geometry with active safety conflicts, infrastructure is the right answer.

O'Hare is a real data point. The O'Hare Modernization Program produced large, durable, measured gains [Contrarian brief, Part I-2]. Any framework that would have deferred this investment needs to explain what would have replaced it. Nothing would have.

NextGen's failure is real. The \$15 billion, 20-year, 16%-of-promised-benefits record is the single hardest fact the operational intelligence case has to answer [Contrarian brief, Part I-5]. The failure was not primarily technical; it was governance, scope creep, and a federal contracting apparatus not built to deliver complex software systems on commercial timelines. But the failure is real, documented, and happened in the exact institutional environment a US operational intelligence expansion would partially inherit.

Safety and depreciation spending is not substitutable. A cracked runway cannot be offset by a surface metering system. The majority of the \$151B ACI-NA need is mandatory compliance [Contrarian brief, Part II, Asset Depreciation]. The 50%+ AIP share for pavement rehabilitation is baseline compliance, not discretionary capacity investment [Contrarian brief, Part II]. This paper's allocation argument does not touch that spend, and Section 3.7 uses the AIP ratio to size the exclusion.

Future demand growth is not refuted by current delay attribution. 1.28B US enplanements by 2038 [Contrarian brief, Part I-4] is a forecast. The 39% late-aircraft share of current delay minutes [Operations brief, Finding 2] is an attribution of *today's* delay. The first is not refuted by the second. The thesis concedes future growth at some airports will be physical. Contrarian Scenario 1 ("Underutilized Hub") and Scenario 3 ("Greenfield Projects in Genuinely Unconstrained Markets") apply in reverse: the thesis is strongest where demand forecasts are fragile or spare capacity exists; weaker where forecasts are both credible and backed by current overload. Denver is the latter [Contrarian brief, Part IV-3]. IAD's international hub thesis is the contested middle.

5.2 Why the thesis still holds

5.2.1 The NextGen indictment is about the wrong category of investment

NextGen modernized the national air traffic control system via a federal program spanning two decades, multiple administrations, and dozens of contractors — reworking a safety-critical distributed system used by every commercial flight in US airspace. Its closest comparables are not airport operational intelligence investments. They are other federal IT modernizations: IRS modernization, DoD defense business systems, the VA electronic health record rollout. Those programs collectively underpin the claim that large federal IT modernizations underperform.

The airport operational intelligence investments this paper argues for differ categorically in three ways.

First, airport-level, not national. A Seattle or Dulles deploying A-CDM operates inside a defined organizational perimeter with an accountable executive. NextGen's stakeholders spanned the FAA, every commercial airline, every controller local, and Congress. Governance complexity compounds rather than adds. The EUROCONTROL A-CDM evidence base — 34,000 tonnes of fuel saved annually across 17 airports, independently compiled [Technology Scout brief, Finding 4] — is what airport-level deployments produce. Not 16% of a promise. An independently audited network-level result.

Second, commercially off-the-shelf. A-CDM, DMAN, computer-vision turnaround monitoring, queue prediction, and surface metering are productized by multiple vendors with multi-year operational track records at European hubs [Technology Scout brief, Sections 2 and 4]. NextGen was custom federal development. The base rate for custom federal IT is not the base rate for commercial airport IT.

Third, scoped in years, not decades. Munich's SESAR DMAN trials ran, produced measured results, and fed into iterative deployment [Technology Scout brief, Finding 4]. Heathrow's A-CDM was live in timeframes orders of magnitude shorter than NextGen's 20-year arc. An airport committing to A-CDM in 2026 knows by 2028 whether it worked. A terminal expansion that breaks ground in 2026 does not reveal its verdict until 2040.

The Contrarian's own Scenario 4 makes the concession: "NextGen's failure reflects 2003-era ambition colliding with 2010s-era technology and 2020s-era implementation capacity. Modern AI capabilities... are categorically more powerful than the rule-based systems NextGen tried to deploy. The historical record does not necessarily indict future capability" [Contrarian brief, Part III, Scenario 4].

5.2.2 The SEA \$13.4M case: a real failure the thesis has to address

The Technology Scout brief states plainly: "Seattle-Tacoma International Airport spent \$13.4 million integrating seven operational intelligence vendors... to achieve basic ramp situational awareness. The case study does not report a single quantified performance outcome. This is the honest picture of what operational intelligence actually costs and how difficult it is to measure" [Technology Scout brief, Quotable 3].

SEA is not NextGen. It is a US, airport-level, commercial-vendor operational intelligence deployment — exactly the category this paper recommends. It produced no published operational results.

V2's "disclosure failure as much as an operational failure" framing was unfalsifiable: by that logic, any future MWAA deployment that failed to publish could be re-characterized the same way — the move the paper criticizes the vendor community for in 4.5. Withdrawn. The rebuttal rests on architecture alone, which is sufficient.

The architecture point: \$13.4M bought seven point solutions, not a coherent platform. The Technology Scout brief identifies the mechanism: "SITA identifies fragmented data as the primary barrier to realizing value from \$8.9 billion in annual airport IT spending. The problem is not insufficient investment in technology. The problem is that the data

infrastructure connecting those investments does not exist" [Technology Scout brief, Quotable 4]. The integration absorbed the investment because the data layer was not built first. SEA must be assumed to have produced no measurable improvement until proven otherwise.

The prescription: a credible MWAA operational intelligence program starts with a unified AODB and data layer, treats point solutions as modules on that layer, and publishes before/after metrics as a contractual condition. Section 6.5 makes that specific. SEA does not invalidate airport-level operational intelligence; it invalidates the "procure seven vendors and integrate at the edges" architecture SEA used.

5.2.3 The Gartner base rate: engaged, not defined around

V1 argued that A-CDM, TBS, and RECAT "are not AI projects" and therefore sit outside the Gartner 28%-success denominator. Special pleading without an independent source. V2 withdrew the claim. V3 refines: the Technology Scout's credibility hierarchy classifies *evidence sources* (EUROCONTROL audits vs. vendor case studies), not *project deployments*. An airport buying A-CDM is not buying from EUROCONTROL — it is buying a deployment from Frequentis, Indra, or Thales, and whether that deployment succeeds depends on local data quality, organizational change, and integration. Those are the exact failure modes Gartner measures.

The honest statement: EUROCONTROL evidence shows that *successfully deployed* A-CDM produces audited fuel and delay savings. It does not show that *attempts* to deploy A-CDM at a new airport succeed at a higher rate than the Gartner base rate. The thesis's claim is:

First, the 28% Gartner figure includes speculative ML projects across every sector of infrastructure and operations broadly. The Technology Scout's hierarchy lets airports select categories with a documented deployment track record (A-CDM at 34 European airports, DMAN in SESAR trials, surface metering at Charlotte). That narrows the failure modes; it does not immunize a deployment from the base rate. The airport buying A-CDM should still expect Gartner-class risk on local integration, data quality, and organizational change.

Second, for categories squarely inside the Gartner denominator — computer-vision turnaround monitoring, queue prediction, predictive maintenance, passenger flow AI — the thesis concedes: expected success rate is lower, data backbone is a prerequisite, and ROI figures should be discounted hard. The Technology Scout brief is blunt: "Every substantive ROI figure in this space requires a credibility haircut" [Technology Scout brief, Finding 3]. Portfolio structure: lead with the strongest track-record categories (A-CDM, DMAN, surface metering — Munich-class numbers), treat mid-confidence vendor-productized categories (turnaround AI, queue prediction) as tier-2 investments sized to fail cheaply, and defer low-confidence categories (operational-simulation digital twins, agentic AI) until the data backbone is demonstrably in place. That structure is what the Gartner data supports.

Third, the comparison case for AI project failure is megaproject failure. Flyvbjerg reports 86% of transport infrastructure projects overrun by 28% on average [Economist brief, Finding 2]. Airport-specific overruns run larger. The right question is not "do AI projects fail?" It is: do airport-level, commercially productized AI projects fail at a higher rate than airport megaprojects, adjusting for magnitude and recoverability? Neither brief answers directly. But the comparison is not infrastructure-reliably-delivers vs. software-reliably-fails. It is: both fail at high rates, with asymmetric financial consequences on airport timescales.

5.2.4 The accountability argument cuts both ways

The Contrarian's claim that infrastructure is accountable while software is not is rhetorically powerful and empirically selective. Denver's DIA Great Hall was inspectable, auditable, and still produced roughly a 3x overrun, a terminated PPP, and a January 2025 city audit finding that DIA "still is not following procurement procedures or requiring contractors to submit documentation of actual costs" [Economist brief, Evidence]. The LAX APM was inspectable, auditable, and has no announced opening date after \$3.34 billion spent against a \$1.9 billion contract [Economist brief, Evidence]. Berlin Brandenburg was the most inspectable airport project of its generation and opened nine years late at roughly €11.9 billion in total value destruction [Economist brief, Evidence].

Infrastructure is visible when complete. During construction — where most of the risk lives — it is not meaningfully more accountable than a software program. The accountability argument is an argument for T5-style governance, not for concrete over code. Section 6.4 operationalizes what T5-class governance means.

5.2.5 The fragility argument is real but asymmetric

The CrowdStrike incident grounded Delta. It did not ground the NAS [Contrarian brief, Part I-6]. The Alaska Airlines hardware failure produced 150 cancellations at a carrier with approximately 1,300 daily flights [UNVERIFIED — HUMAN REVIEW — 1,300 daily flights figure is analyst context, not brief-sourced] — a bad day for the carrier, not an NAS-level systemic failure. Software fragility is real and has to be designed around. But the asymmetric case cuts the other way too: a runway that cannot be built cannot ground flights it was not yet serving. A terminal canceled at planning does not trap 30 years of capital. An airport-level operational intelligence failure costs tens of millions; a failed terminal costs billions and encumbers the balance sheet for decades.

The correct response to software fragility is redundancy, backup protocols, and failover design — at a fraction of physical-redundancy cost. Not concrete.

5.2.6 Which US hubs the thesis applies to — with names

The v1 draft asserted the thesis conditions "describe a supermajority of the US large-hub system." That quantified claim is not supported in any Stage 1 brief, and "multiple" without names is hand-waving. V3 names the hubs.

Five conditions: (a) peak-hour utilization does not sit above 90% of declared capacity for the bulk of operating hours; (b) delay attribution shows meaningful contribution from air carrier, late aircraft, ATC staffing, or ground coordination; (c) the airport has not deployed core commercially productized operational intelligence categories at scale; (d) the capital program's debt trajectory is moving toward DSCR below 1.5x within five years; (e) demand forecasts embed assumptions that could be disrupted by hub-strategy changes, fuel-price shocks, or post-pandemic pattern shifts.

Condition (c) is the one Stage 1 evidence addresses most directly: only 3 of 31 US large hubs have operational digital twins, with 17 of 31 in pilot or planning [Technology Scout brief, Finding 2]; A-CDM at US-hub scale does not exist [Technology Scout brief, Finding 4]; SFO's AIOC was one of the first serious US APOC deployments as of January 2026 [Technology Scout brief, Section 4]. By (c), roughly 28 of 31 US large hubs qualify.

(a), (b), (d), (e) require airport-by-airport verification. Candidate hubs [UNVERIFIED – HUMAN REVIEW – analyst-identified candidates; conditions (a) and (d) not verified for each hub]:

- **ATL**: largest US hub by movements; not at hard physical cap; substantial Delta coordination and late-aircraft exposure; IAD peer on hub-concentration risk. (c) likely applies; (b) and (e) plausibly apply; (a) and (d) require filings.
- **MCO**: growing non-hub O&D traffic; physical room to expand; high ground coordination variance (OAG cites Orlando's JetBlue 25% turn variance) [Operations brief, Finding 3]. (b), (c), (e) plausibly apply.
- **IAH**: United hub with network-decision risk analogous to IAD's; significant ramp operations complexity. (b), (c), (e) plausibly apply.
- **DTW** or **MSP**: Delta hubs with CVG-adjacent network-risk exposure [Economist brief, Finding 7]. (b), (c), (e) plausibly apply.
- **IAD**: United hub with documented DSCR deterioration to 1.70x by FY2030 [Economist brief, Finding 5]; (d) directly applies; (b), (c), (e) plausibly apply.

Five candidate US large hubs where the thesis plausibly applies. Neither brief enumerates the exhaustive list. (a) and (d) require airport-specific disclosures neither Stage 1 brief compiles. Human reviewer should verify the five-condition test against each hub's published data before citation outside this paper. The thesis's analytical scope is the set of hubs passing the test; its MWAA application is Section 6.

5.3 The honest disposition of the argument

Concede: genuine physical ceilings; safety-and-depreciation spending as the dominant share of the \$151B ACI-NA need; O'Hare-class legitimate infrastructure cases; NextGen as a real warning about federal IT governance; SEA as a real warning about airport-level integration architecture; the Gartner base rate as a real portfolio-selection constraint; and the FAA/IATA forecast of sustained demand growth. Do not concede: the aggregate allocation ratio on the discretionary tranche; the false equivalence between federal-scale modernization and airport-level commercially productized operational technology; the inference that *current* delay in 2026 is primarily a physical-capacity problem — 39% of

delay minutes come from late-arriving aircraft and 40% of NAS delays come from 19 understaffed ATC facilities, and those attributions are not overturned by the forecast. The case is not against all concrete. It is against unexamined discretionary concrete, and for treating the operational intelligence stack as the first option to defeat rather than the last.

6. Implications for MWAA

MWAA runs an airport at a federal-and-physical cap (DCA) and an airport with physical room but exposed hub economics (IAD). The allocation argument applies to both, in different forms. This section names specific projects, operational-intelligence complements, investment envelopes, KPIs, and defeat tests. V2's envelopes over-attributed to the Technology Scout brief; v2's defeat tests threatened only the operational intelligence investment, not any infrastructure decision. V3 resolves both.

6.1 DCA: the physical-ceiling case, with nested constraints

DCA has two binding constraints. A: physical geography (Potomac, federal facilities, residential neighborhoods) plus a terminal designed for 15M annual passengers now serving 25M [Contrarian brief, Part I-1; Part II]. B: federal policy — a 62-per-hour slot cap plus the perimeter rule [Contrarian brief, Part II]. A is physical and not movable. B is policy — theoretically movable but has not moved despite decades of political pressure.

The \$2.4 billion DCA Project Journey program (Terminal B/C redevelopment within the 2025–2039 plan) responds to Constraint A [Economist brief, Evidence: MWAA-Specific]. It does not add flights; it handles passengers associated with flights DCA is already allowed to operate. Placed straight, this framing puts *all* of Project Journey in the non-discretionary tranche and makes the DCA defeat test a test only of the operational intelligence investment — inverting the thesis. V3 splits Project Journey into a non-discretionary core and a discretionary sub-tranche, applies the defeat test only to the latter, and attaches an infrastructure consequence to a named operational outcome.

Project Journey non-discretionary core: terminal roof and structural renewal; passenger-access rework tied to federal accessibility standards (ADA and TSA); security checkpoint footprint to meet federal requirements; baggage system replacement at end-of-life; apron-level pavement tied to FAA PCI compliance. Proceeds on its own terms; the allocation argument does not touch it.

Project Journey discretionary sub-tranche: gate-count additions beyond what the 62-per-hour slot cap requires (gates sized for schedule peaking rather than slot-cap saturation); hold-room square footage beyond federal code minimums; concession and retail build-out; passenger-experience amenities (premium lounges, wayfinding beyond accessibility minimums). Human reviewer should establish the dollar value from the authority's capital plan. Analyst estimate: 15–30% of \$2.4B — **\$350–750M** [UNVERIFIED — HUMAN REVIEW]. If 70–85% of Project Journey is structural, accessibility, security, baggage, and pavement (by analogy to the AIP >50% pavement

ratio plus non-pavement mandatory categories at a terminal redevelopment), then 15–30% is the residual. Analyst inference, not brief-anchored; verify against capital plan line items before external citation.

Operational intelligence complement (coupled program, not a serially-following IT subline):

- (i) Security queue prediction and staffing optimization (Copenhagen Optimization / Xovis / Vision Platform AI) [Technology Scout brief, Queue Prediction].
- (ii) Biometric passenger processing at gates and immigration where federal rules allow [Technology Scout brief, Biometric Passenger Flow].
- (iii) Predictive gate resource allocation and ground coordination including computer-vision turnaround monitoring [Technology Scout brief, Predictive Turnaround/Ramp Coordination].
- (iv) An airport operations data backbone (unified AODB, real-time feeds, data-sharing agreements) making (i)–(iii) coherent [Technology Scout brief, Quotable 4].

Indicative investment envelope — analyst bottom-up from component-level brief figures:

The Technology Scout brief's only explicit large-hub envelope is for digital twin deployments (\$15–40M over 24–36 months, plus 10–20% annual recurring) [Technology Scout brief, Evidence: Digital Twins]. There is no brief-level cost figure for an integrated A-CDM + DMAN + surface metering + queue prediction + turnaround AI stack at a US large hub. The envelopes below are analyst-derived from brief-level component anchors.

Component anchors:

- Digital twin large-hub deployment: **\$15–40M** over 24–36 months, 10–20% annual recurring [Technology Scout brief, Evidence: Digital Twins].
- Digital twin PoC: **\$150K–\$200K**; pilots \$50K–\$200K. DFW's 2022 Willow/Parsons contract: **~\$2.9M** initial scope, one runway and one terminal [Technology Scout brief, Digital Twins].
- SEA's seven-vendor ramp integration: **\$13.4M** without unified data backbone (poor-architecture counterexample, not a price benchmark for well-architected systems) [Technology Scout brief, Finding 1; Quotable 3].
- Queue prediction (Copenhagen Optimization and peers): no explicit capital envelope; productized, deployable in months, documented at 40+ airports [Technology Scout brief, Queue Prediction].
- Biometric passenger flow: hardware mature; no airport-level cost envelope in brief [Technology Scout brief, Biometric Passenger Flow].
- Computer-vision turnaround monitoring (Assaia-class): productized at 15+ major airports; no airport-level cost envelope in brief; Heathrow's deployment scaled to 116 gates across three terminals [Technology Scout brief, Predictive Turnaround/Ramp Coordination].

DCA stack envelope, analyst-derived: The four-category stack (queue prediction + biometric passenger flow + computer-vision turnaround + data backbone) is not a digital twin at large-hub scale, but its integration complexity and data-backbone dependency put it in the same order of magnitude. Anchored to the digital twin envelope and informed by SEA's cautionary integration cost: **\$15–50M over 24–36 months plus 10–20% annual recurring** [UNVERIFIED – HUMAN REVIEW]. Low end: three productized modules on a focused data layer (closer to the \$15M floor). High end: integration closer to the \$40M ceiling plus modest additional vendor cost. Analyst judgment; human reviewer should verify against MWAA procurement records at comparable hubs if available before external citation.

KPIs with disclosed baselines (the SEA lesson):

- Security peak-hour wait time (median and 95th percentile) — measurable reduction within 12 months; Copenhagen Optimization's 30–50% peak-wait reduction across 40+ airports is the vendor benchmark, discounted toward Munich-class low-double-digit range for realistic ambition [Technology Scout brief, Queue Prediction].
- Security throughput per FTE per hour — directional movement toward Copenhagen's 42% improvement, with a disciplined discounted target set pre-deployment [Technology Scout brief, Queue Prediction].
- Taxi-out time median per departure — Munich-class 9% reduction over 24 months [Technology Scout brief, Finding 4].
- On-time departure performance, stripping schedule padding per 3.9, with airline agreement on the adjusted baseline.

Defeat test with a named infrastructure consequence. The discretionary sub-tranche (\$350–750M; authority filings to govern the number) is held in reserve pending the operational outcome. *The formulas below are **proposed pre-registered contract terms**, not existing MWAA policy:*

- If by month 24, security peak-hour 95th-percentile wait time has fallen $\geq 20\%$ sustained over six consecutive months from a disclosed baseline, the discretionary sub-tranche — specifically the gate-count and hold-room components sized for schedule peaking — is **resized downward by a formula pre-registered at contract signature**: for every 10% sustained 95th-percentile wait reduction, one increment of discretionary hold-room square footage is deferred or cancelled. [UNVERIFIED – HUMAN REVIEW – proposed mechanism; "increment" requires definition in the actual contract]
- If by month 24 the Munich-class 9% taxi-out reduction is achieved (sustained six months), the apron-side discretionary elements are **deferred 36 months** pending a repeat measurement.
- If neither threshold is met, the operational program pauses for independent post-mortem; the discretionary sub-tranche proceeds on original terms only after an independent review identifies the failure mode. The non-discretionary core proceeds regardless.

- Trigger thresholds (20% 95th-percentile wait reduction; 9% taxi-out reduction) sit below the Copenhagen vendor benchmark and at the Munich single-airport benchmark, respectively. Achievable for a well-executed deployment; not achievable by noise.
- Adjudicator: a third-party operational-audit firm contracted at the start — not the authority, not the vendor. Naming the firm is a governance decision, not a Stage 2 analytical one.

The architecture lesson from SEA: the data backbone (unified AODB, real-time feeds from airlines and ground handlers, documented data-sharing agreements) is the first investment, not the seventh. SEA spent \$13.4M integrating seven vendors without that backbone and produced no disclosed outcome [Technology Scout brief, Quotable 3]. DCA should not repeat that architecture.

6.2 IAD: the demand-exposed hub

Dulles has physical room. It has an active Concourse E expansion within the \$9 billion 2025–2039 program [Economist brief, Evidence: MWAA-Specific]. It is a primary international gateway with legitimate long-haul hub economics. It is also the exposed flank of MWAA's balance sheet: DSCR projected to deteriorate from 2.79x in FY2024 to 1.70x by FY2030 even under favorable traffic assumptions [Economist brief, Finding 5], and system-level adjusted debt per O&D enplanement peaking near \$400 in 2028 implicates Dulles most.

The CVG warning applies to IAD more than DCA. CVG's CPE rose from under \$5 to over \$10 on collapsing enplanements after Delta dehubbed [Economist brief, Finding 7]. IAD is not CVG, but it is more structurally exposed to airline network decisions than DCA. A capital program sized to a traffic forecast that assumes continued hub commitment from United carries asymmetric risk: if the hub holds, the investment earns; if it thins, the investment encumbers the balance sheet with bond obligations that follow the Authority regardless of who is flying.

Project and staging. IAD Concourse E expansion within the \$9B program. Stage 1 briefs do not disclose a Concourse E sub-project cost; the Economist brief cites WJLA coverage of the \$9B authority-wide contract with Concourse E and Project Journey both included [Economist brief, Evidence: MWAA-Specific], without decomposition. Stage 3 should establish the figure from the published plan. For now:

- **Concourse E high-confidence core tranche:** airside infrastructure, apron, and taxiway work tied to aircraft-size/NextGen compatibility, code compliance, and asset renewal. Proceeds on its own financial case.
- **Concourse E demand-contingent tranche:** gate count beyond current scheduled-movement peaks, hold room expansion, and terminal square footage justified by forecast international growth. Subject to the defeat test below. Analyst estimate: **\$200M–\$600M** [UNVERIFIED – HUMAN REVIEW], subject to verification against MWAA's published capital plan. (Sub-concourse components of a US

international-gateway expansion typically run in the low-to-mid hundreds of millions per demand-contingent tranche; analyst judgment.)

Operational intelligence complement: A-CDM / DMAN paired with DCA (see below), surface metering (TFDM-class), computer-vision turnaround monitoring at high-variance stands, and an APOC container (see 6.5). Scoped to move IAD's throughput baseline before the demand-contingent tranche is committed.

The network-effect resolution. V2 transmitted the A-CDM network-effect caveat in 3.4 but sized IAD recommendations against Munich's numbers without resolving the contradiction. V3 resolves it: MWAA should deploy A-CDM at *both* DCA and IAD as a paired regional cluster. Justified because (a) MWAA operates both airports under a single authority and can commit to a paired deployment a single-airport authority cannot, (b) a two-airport cluster captures some but not all of the European network effect, and (c) MWAA can lobby the FAA and BWI to form a regional cluster that captures more. The cost envelope accommodates the paired deployment; the KPI benchmark is Munich's single-airport figures as floor rather than the 17-airport EUROCONTROL network figures as ceiling.

Indicative envelope for IAD — analyst bottom-up: Same component-anchored construction as 6.1, plus an A-CDM/DMAN component sized against Munich's single-airport benchmark. The brief provides no explicit A-CDM cost envelope for a US single airport; Munich's SESAR DMAN was part of a multi-year research program with undisclosed installation cost. By analogy to the digital twin \$15–40M benchmark and accounting for A-CDM being more data-governance-heavy than software-license-heavy, the IAD envelope lands at approximately **\$20–60M over 24–36 months plus 10–20% annual recurring** [UNVERIFIED — HUMAN REVIEW]. Analyst-derived; not a brief-cited A-CDM cost. Human reviewer should attempt to anchor to Frequentis, Indra, or Thales published deployment costs at comparable hubs before external citation.

KPIs with disclosed baselines:

- Taxi-out time at IAD (median and 95th percentile per departure) — Munich-class **9% reduction** as 24-month ambition, used as defeat-test floor rather than the EUROCONTROL 17-airport aggregate [Technology Scout brief, Finding 4].
- Off-block time predictability — Munich-class **7.8% improvement** [Technology Scout brief, Finding 4].
- Turnaround delay median — Assaia-class 1-minute reduction as ambition, with the vendor-sampling caveat transmitted [Technology Scout brief, Finding 7, Section 4]; defeat-test floor is a more conservative 0.5-minute reduction given the caveat.
- Peak-hour aircraft movements per runway per hour — baseline before A-CDM, tracked monthly after.
- Internal tracking of whether IAD's isolated-deployment gains underperform the 17-airport EUROCONTROL benchmark, with a pre-registered acceptance criterion that Munich is the floor and the 17-airport network benchmark is not.

Defeat test with a named infrastructure consequence. V2's IAD defeat test created a perverse incentive (operational intelligence failure was the only path to the demand-contingent tranche proceeding unmodified) and "by measured amounts" was undefined. V3 specifies the thresholds numerically and eliminates the perverse-incentive path. *The formulas below are **proposed pre-registered contract terms**, not existing MWAA policy:*

- If by month 24, taxi-out median at IAD has fallen by $\geq 7\%$ sustained over six consecutive months from a disclosed baseline, **and** peak-hour movements per runway per hour have increased by $\geq 4\%$ sustained over six months, the Concourse E demand-contingent tranche is **resized downward by a formula pre-registered at contract signature**: demand-contingent gate count is reduced proportionally to the throughput gain (7% throughput gain yields at least 3.5% gate-count reduction; 10% yields 5%). [UNVERIFIED – HUMAN REVIEW – proposed mechanism; coefficient is analyst choice, not empirically derived] Real infrastructure consequence conditioned on a named operational outcome. Thresholds sit below Munich's 9% and 7.8% benchmarks to be achievable for a well-executed US-isolated deployment.
- If the thresholds are *not* met, the demand-contingent tranche does **not** automatically proceed. An independent adjudicator evaluates whether the failure was architectural (SEA-class), execution (data backbone, vendor, integration), or environmental (genuine physical ceiling). Architectural or execution: restructure the operational intelligence program, defer the tranche 18 months for a repeat measurement. Environmental (well-executed program, genuine throughput ceiling): the tranche proceeds on original terms. No path lets operational intelligence failure alone unlock the infrastructure spend; the failure must be adjudicated, and environmental failure — the only unlocking path — is also the finding that most justifies the infrastructure spend on the merits.
- The adjudicator is a third-party contracted at the start of the operational intelligence program. The authority is not both player and referee.

6.3 The financial envelope question, reconciled

The authority is at a specific decision point. The 2025–2039 plan is approximately 5.6x the prior cycle; \$5.5 billion of new debt is scheduled for 2025–2028; Moody's sector outlook is negative; the PFC cap is frozen at \$4.50 [Economist brief, Finding 5, Finding 3]. The marginal bond is more expensive, more scrutinized, and more consequential than at any point in the authority's recent history.

Reconciled system-level envelope. Combining DCA from 6.1 (\$15–50M), IAD from 6.2 (\$20–60M), and APOC from 6.5 (\$15–40M combined), the system-level operational intelligence band is **\$50–150M over 36 months, plus 10–20% annual recurring** [UNVERIFIED – HUMAN REVIEW – analyst sum of analyst-derived components]. This is the number Section 6.6 reconciles to. Prior drafts referenced "\$100M–\$150M" and a separately constructed \$55–135M figure that did not reconcile to the component sums; both are superseded. Midpoint approximately \$100M.

The allocation should not be framed as a subtraction from infrastructure. It is an uplift to the revenue base that supports the infrastructure. A Munich-class 9% taxi time reduction [Technology Scout brief, Finding 4] or a EUROCONTROL-class 0–3 minute per-departure taxi-out saving across MWAA's annual operations translates into fuel savings, gate availability, and delay reductions that feed both airline economics (shaping CPE tolerance) and passenger experience (shaping non-aeronautical revenue).

Sizing: \$50–150M over 36 months is approximately **0.6–1.7% of the \$9B capital plan** and on the order of **2–15% of the \$1–3B discretionary tranche from Section 3.7**. Not the heroic reallocation the shorthand suggests. A single-digit-percentage allocation against the discretionary tranche that buys a defeat test on specific named sub-projects worth an analyst-estimated **\$550M–\$1.35B** combined (DCA discretionary \$350–750M plus IAD demand-contingent \$200–600M). Ratio: approximately **1:6 to 1:9** — large enough to be materially consequential if the operational intelligence succeeds, small enough to be a rounding error if it does not.

6.4 Governance: what "T5-class" actually means

The Stage 1 briefs describe T5's "two-year pre-construction study explicitly designed to avoid megaproject failure modes" [Economist brief, Evidence: Cost Overrun Case Studies] but do not detail the governance mechanisms. Rather than invoke a phrase the briefs do not unpack, this paper substitutes failure-mode controls derived from the Denver, LAX APM, and Berlin cases:

- **Ring-fenced scope at contract signature.** Post-signature changes require board-level approval with disclosed cost impact. Denver Great Hall's PPP terminated because cost-and-scope disputes were not managed at that level [Economist brief, Evidence].
- **Independent cost validation before each major gate.** Not contractor-provided, not owner-provided — third-party validation, published to the board.
- **Contractual clarity on risk allocation.** Which party bears escalation, design-change, weather-delay, and regulatory-change risk is specified at signature and does not migrate.
- **Contingency reserve discipline.** Contingency is ring-fenced at project level and cannot be cross-subsidized without explicit board action.
- **Procurement discipline audited externally.** The DIA 2025 audit finding — "still is not following procurement procedures or requiring contractors to submit documentation of actual costs" [Economist brief, Evidence] — is the exact failure mode MWAA cannot replicate on a 5.6x program. Annual independent procurement audit published to the board.

MWAA's \$9B program at the 5.6x step-up needs its governance architecture stress-tested by someone other than the people who wrote it. A specific audit and a specific set of procurement controls.

6.5 The APOC question: the container, not the room

SFO opened the first serious US Airport Operations Center in January 2026 — a 22,000-square-foot facility with 67 workstations [Technology Scout brief, Section 4].

Documented operational results are not yet published. A container. Not yet a capability.

EUROCONTROL's April 2025 APOC Performance Study reviewed European APOC implementations, surveyed 62 participants from 34 airports, documented a catalogue of performance indicators and "reported benefits" — and did not quantify those benefits with audited before/after comparisons [Technology Scout brief, A-CDM and DMAN]. After years of European APOC deployment, rigorous cross-airport performance comparison still does not exist in the literature. The European APOC benchmark is less evidence-rich than A-CDM or DMAN, and commitments should be sized accordingly.

MWAA has neither a DCA nor an IAD APOC in the European sense. The question is not whether to build one. The question is whether the organizational, data, and governance architecture that makes an APOC coherent exists — a unified AODB, real-time feeds from major operational stakeholders, formal data-sharing with airlines and ground handlers, and accountable executive ownership of operational outcomes. Without that architecture, an APOC is a room with screens. With it, it is the container that makes every other operational intelligence investment compound.

Prerequisite steps, before the room:

- Internal audit of current AODB coverage and real-time feed completeness at both airports, published to the board.
- Data-sharing agreements with the five largest airlines and major ground handlers, with agreed latency and completeness SLAs.
- A named accountable executive — not a committee — for operational performance outcomes at each airport.

APOC envelope — analyst bottom-up: The brief provides no cost envelope for an APOC facility or its underlying data platform; SFO's 22,000-square-foot, 67-workstation facility is described without a dollar figure [Technology Scout brief, Section 4]. Using the digital twin benchmark as the closest anchor (\$15–40M large-hub operational simulation) plus a conservative allowance for physical facility and AODB data-backbone build, the envelope is approximately **\$15–40M combined across both airports** over 24–36 months, staged behind the data-backbone prerequisite. [UNVERIFIED — HUMAN REVIEW] Analyst-derived; not a brief-cited APOC cost.

- KPI: taxi-out median, gate-availability predictability, on-time departure (padding-adjusted), weather-event recovery speed.
- **Defeat test with a named infrastructure consequence.** If 18 months after data-backbone deployment the KPIs have not moved from a disclosed baseline by Munich-class magnitudes (9% taxi-out reduction minimum, 5% gate-availability-predictability improvement minimum), the APOC facility investment at DCA is **deferred indefinitely** pending independent operational review, and the residual

facility budget is reallocated either to the Project Journey non-discretionary core (if DCA) or the Concourse E high-confidence core (if IAD). This conditions a physical-facility commitment (the APOC room and workstations themselves) on a named operational outcome — the thesis v2 failed to implement in Section 6's defeat tests.

6.6 What this piece is not

It is not a recommendation to pause Project Journey, halt Concourse E, or cancel any committed project. Pavement, asset renewal, code compliance, and capacity components responsive to documented demand are legitimate investments. The thesis is narrower than v1 implied:

The discretionary tranche of MWAA's \$9B capital plan — analyst-estimated at \$1–3B `[UNVERIFIED — HUMAN REVIEW]`, and specifically the Project Journey discretionary sub-tranche (analyst-estimated \$350–750M) `[UNVERIFIED — HUMAN REVIEW]` and the Concourse E demand-contingent tranche (analyst-estimated \$200–600M) `[UNVERIFIED — HUMAN REVIEW]` — should have to pass a defeat test against a \$50–150M operational intelligence alternative `[UNVERIFIED — HUMAN REVIEW]` deployed first, measured against disclosed-baseline KPIs with Munich-class minimum thresholds, with published outcomes, named sub-project consequences pre-registered at contract signature, and independent adjudication.

That thesis is the same across Sections 1, 3.10, and 6.6.

The bond market, the rating agencies, and the authority's own DSCR trajectory now require that question be answered on the record, not assumed away. The next approximately \$1–3B of discretionary MWAA CapEx is going to get decided. It should be decided knowing what **\$50–150M** of operational intelligence — approximately one-sixth to one-ninth of the discretionary tranche it conditions — would have done first, and knowing the answer is not allowed to be "we didn't ask."

Draft v3 Editor's Cut. Inline citations reference the Stage 1 briefs. Dollar envelopes in Section 6 are analyst-derived from Technology Scout benchmarks (digital twin \$15–40M large-hub; DFW \$2.9M partial-pilot; SEA \$13.4M poor-architecture counterexample) and flagged for Stage 3 verification against MWAA procurement records and authority filings. Project Journey and Concourse E sub-tranche sizing is analyst construction from the AIP mandatory-ratio anchor and brief-level project identification; requires Stage 3 verification against MWAA's published capital plan line items.

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